Exercise 8

$$U(x - \Delta x) = U(x) - \left(\frac{\partial U}{\partial x}\right)\Delta x + O(\Delta x^2)$$

$$\frac{U(x - \cancel{\Delta x})}{\cancel{\Delta x}} = \frac{U(x) - \left(\frac{\partial U}{\partial x}\right)\cancel{\Delta x} + O(\cancel{\Delta x}^2)}{\cancel{\Delta x}}$$

$$\frac{\partial v}{\partial x} \; O(\Delta x) = \frac{-[v(x - \Delta x) - U(x)]}{\Delta x}$$

$$\frac{\partial v}{\partial x} = -\frac{[v(x - \Delta x) - v(x)]}{\Delta x} + O(\Delta x)$$

$$\frac{\partial v}{\partial x} \approx -\frac{[v_{i+1} - v_i]}{\Delta x}$$

Exercise 9

- $x_i = x_0 + i(\Delta x)$
- $t^n = t_0 + n(\Delta t)$

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$$\frac{\partial^2 U}{\partial x^2} = \frac{U(x+\Delta x) - 2U(x) + U(x_i - \Delta x)}{\Delta x^2} + \theta(\Delta x^2)$$

$$\frac{\partial^2 U}{\partial x^2} = \frac{U(x_{i+1}) - 2U(x_i) + U(x_{i-1})}{\Delta x^2} + \theta(\Delta x^2)$$

$$\frac{\partial^2 U}{\partial x^2} = \frac{U_{i+1} - 2U_i + U_{i+1}}{\Delta x^2} + \theta(\Delta x^2)$$

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$$\frac{\partial U}{\partial z} = \frac{U(z_i + \Delta z) - U(z_i - \Delta z)}{2\Delta z} + \theta(\Delta z^2)$$

$$\frac{\partial U}{\partial z} \approx \frac{U_j^n(k+1) - U_j^n(k-1)}{2\Delta z}$$